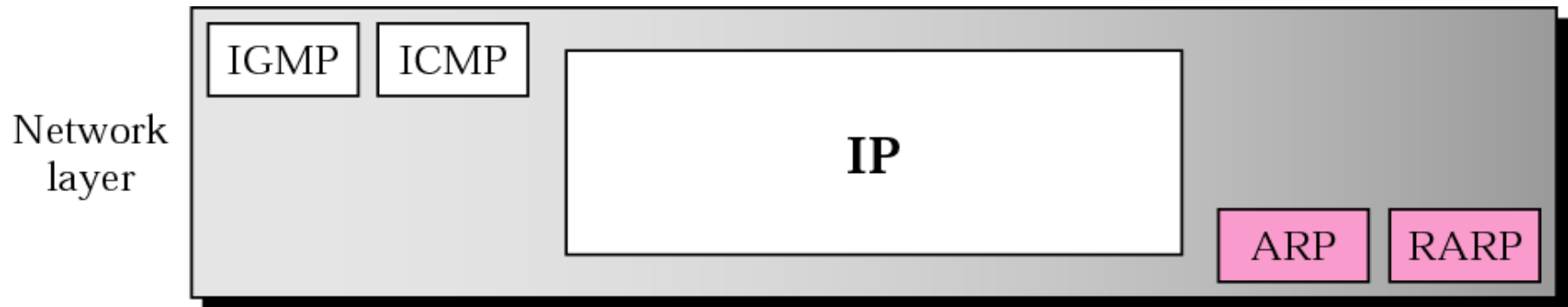
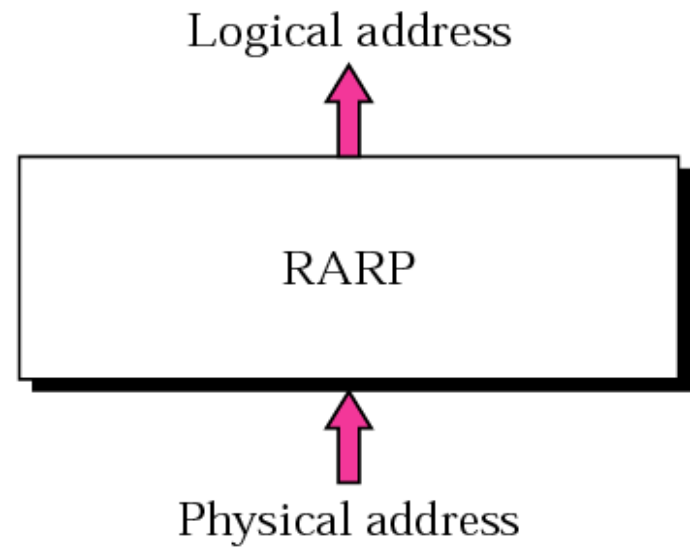
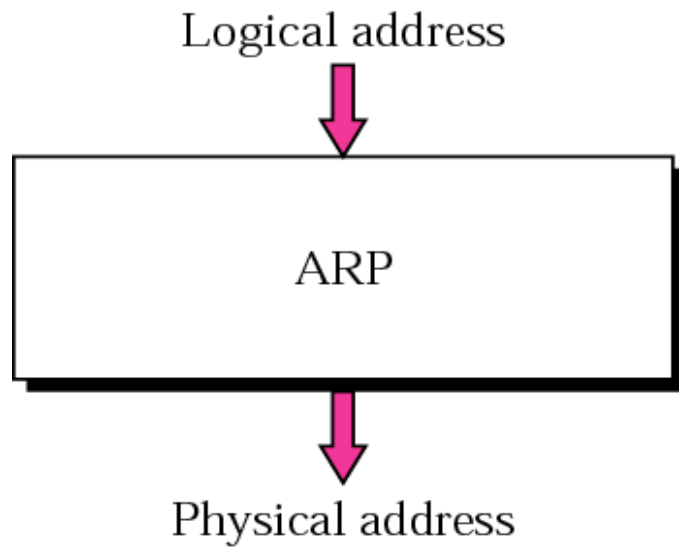


# Network Layer Protocols

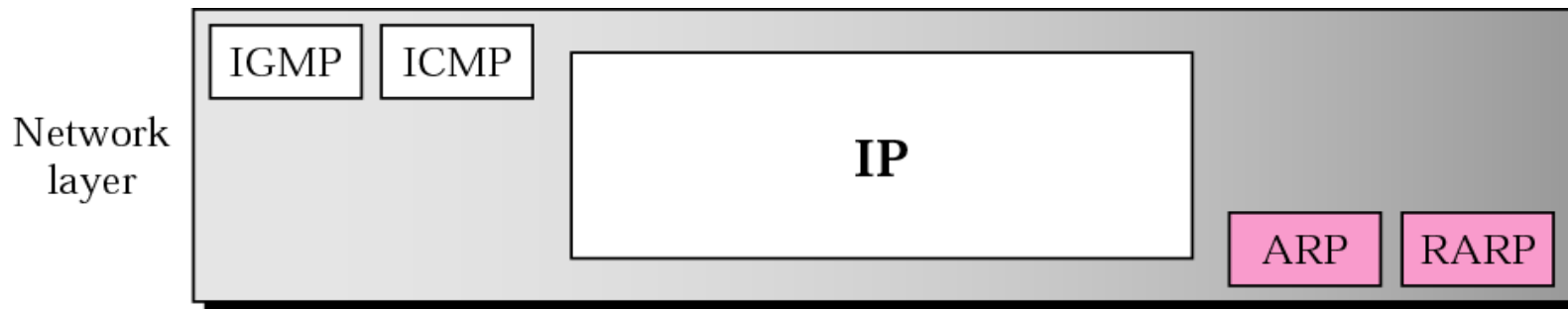




# *ARP and RARP*



**Figure 7.2** *Position of ARP and RARP in TCP/IP protocol suite*



**Notice that ARP and RARP are supplemental to IP.**

# 7.1 ARP

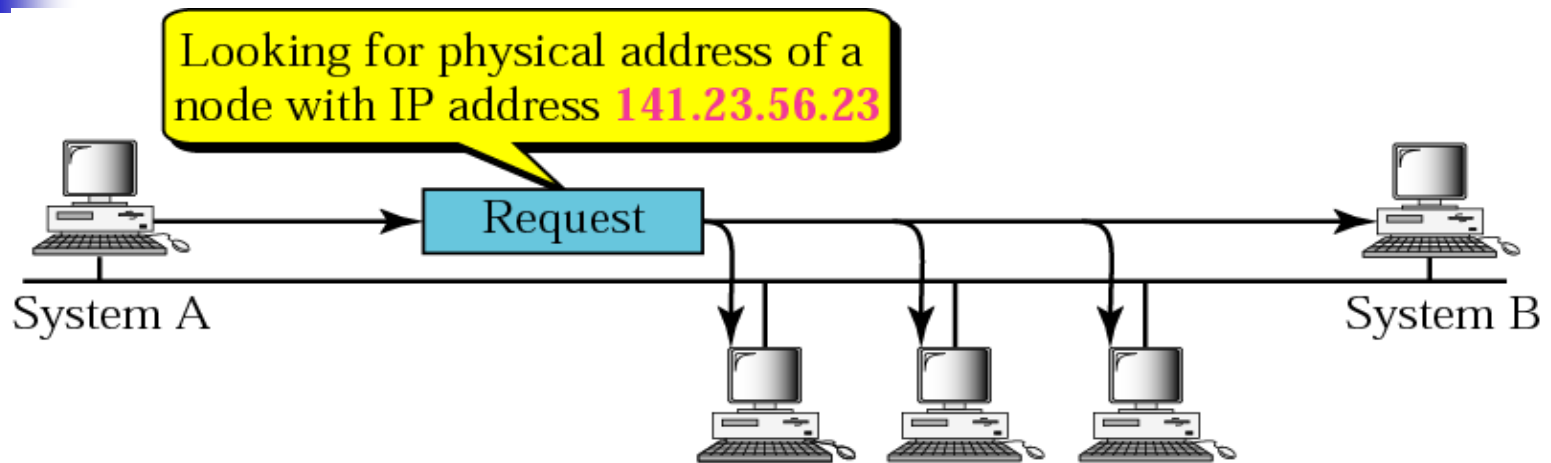
- Node having datagram to send has IP address of receiving node. Last router in the path knows the IP address of destination host but it is not used to move a frame through link because MAC address is needed. Here ARP is required.
- ARP associates an IP address with its physical address. On a physical network, such as a LAN, each device on a link is identified by a physical or station address that is usually imprinted on the NIC.

# ADDRESS MAPPING

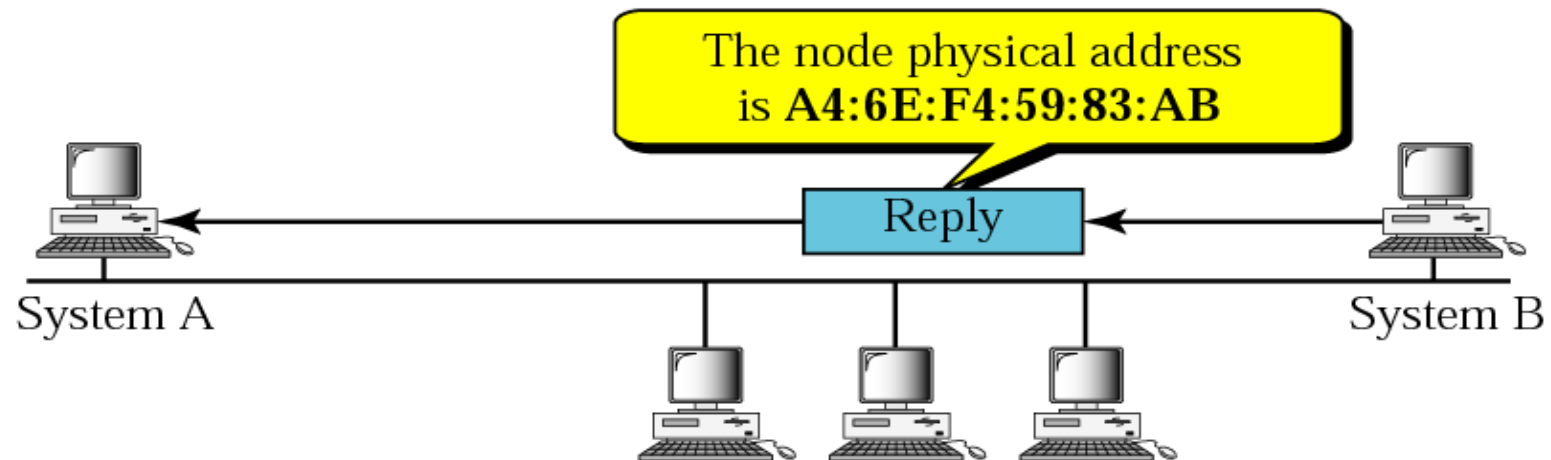
The delivery of a packet to a host or a router requires two levels of addressing: **logical** and **physical**. We need to be able to map a logical address to its corresponding physical address and vice versa.

**Logical address to physical address translation can be done statically (not practical) or dynamically (with ARP).**

**Figure 7.3** *ARP operation*

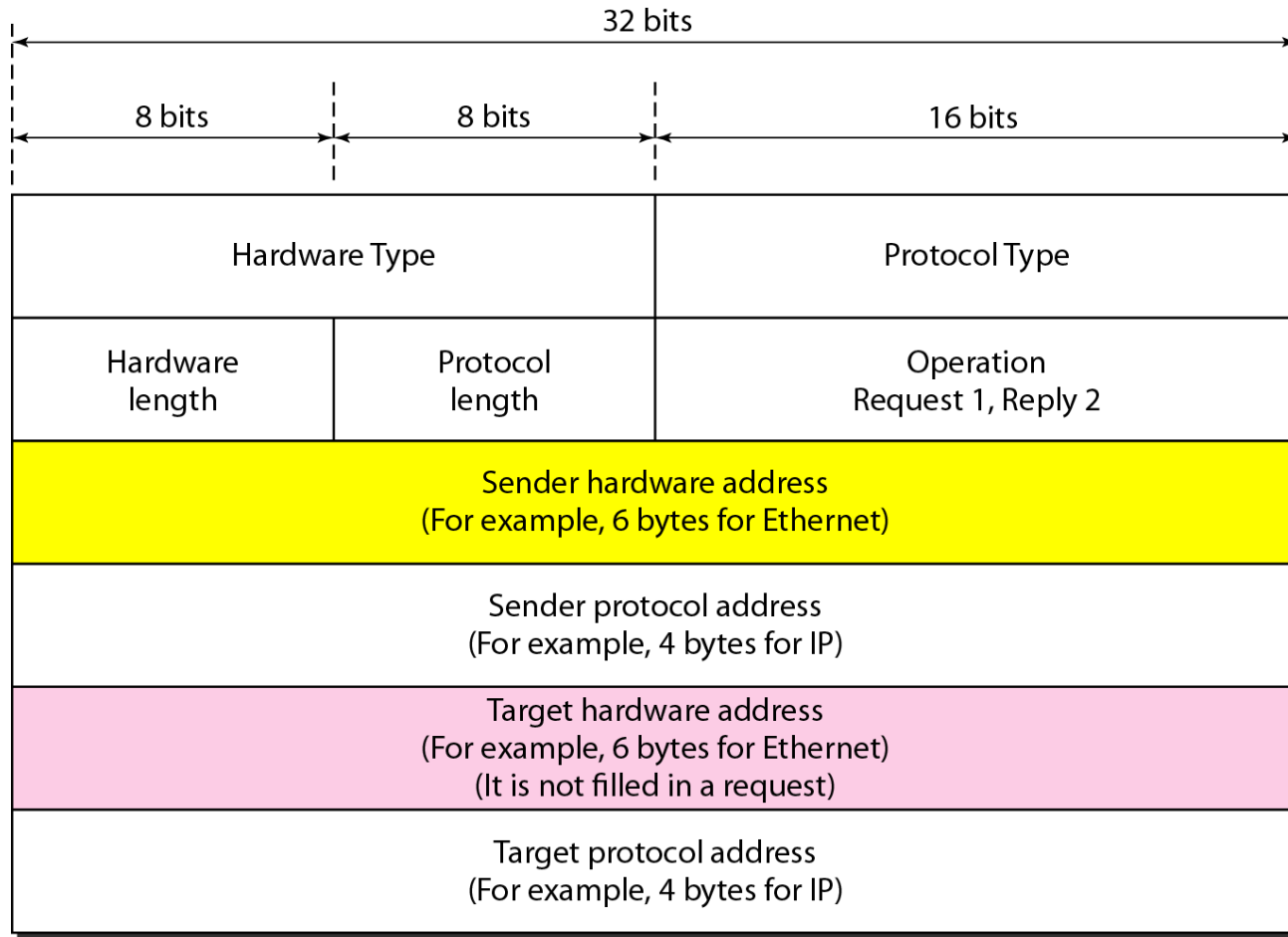


a. ARP request is broadcast



b. ARP reply is unicast

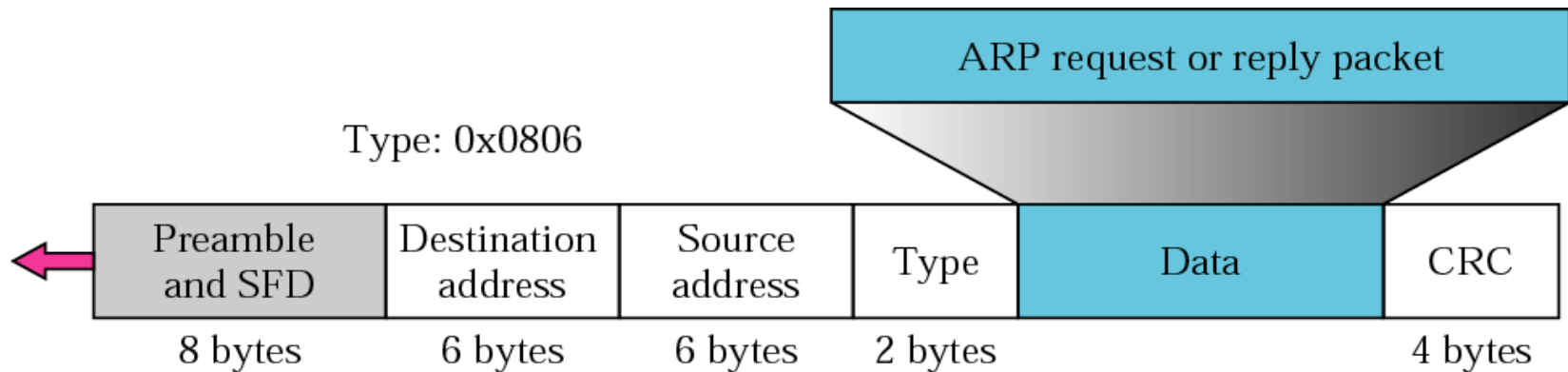
**Figure 21.2** *ARP packet*



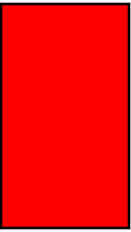
- **Hardware type (16-bit)** field defines the type of link layer protocol (Eg. Ethernet).
- **Protocol type (16-bit)** field defines the network layer protocol (IPv4).
- **Hardware Length (8-bit)** field defining the length of the physical address in bytes. For example, for Ethernet the value is 6.
- **Protocol Length (8-bit)** field defining the length of the logical address in bytes. For example, for the IPv4 protocol the value is 4.
- **Operation (16-bit)** field defining the type of packet. Two packet types are defined: ARP request (1) and ARP reply (2).



**Figure 7.5** *Encapsulation of ARP packet*



**The ARP packet is encapsulated within an Ethernet packet.  
Note: Type field for Ethernet is x0806**

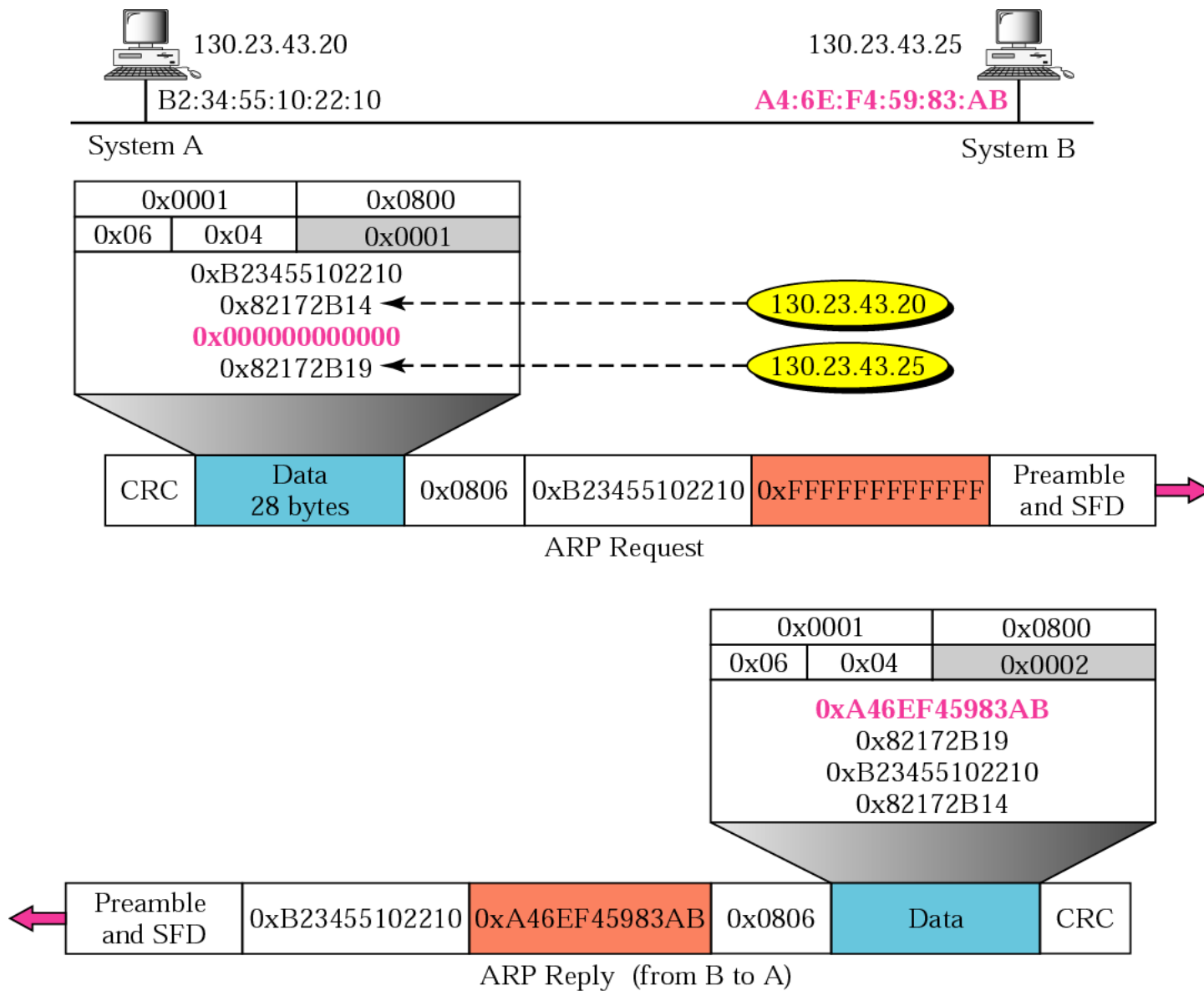


## ***EXAMPLE 1***

*A host with IP address **130.23.43.20** and physical address **B2:34:55:10:22:10** has a packet to send to another host with IP address **130.23.43.25** and physical address **A4:6E:F4:59:83:AB** (which is unknown to the first host). The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.*

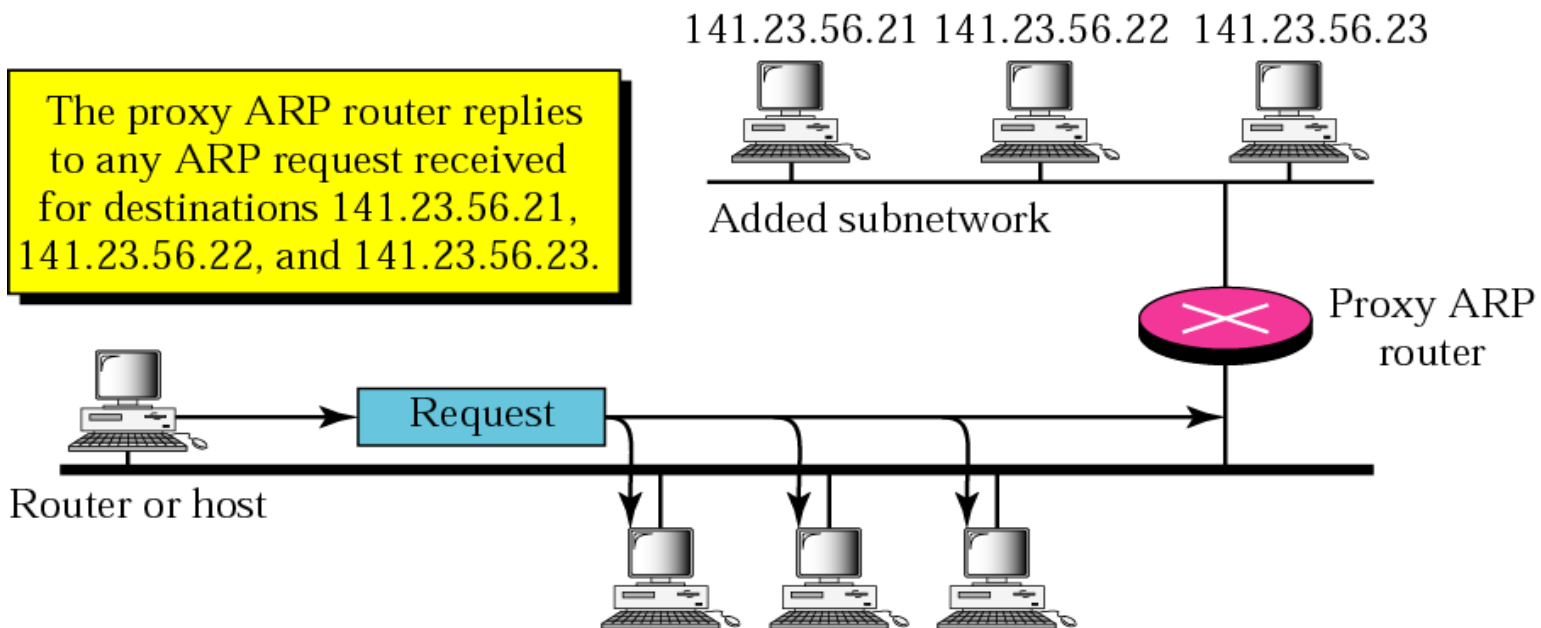
**See Next Slide**

**Figure 7.7 Example 1**



**Figure 7.8** *Proxy ARP*

- A Technique called proxy ARP is used to create a subnetting effect.
- Proxy ARP acts on behalf of a set of hosts.
- A proxy ARP, running in a router, can respond to an ARP request for any of its hosts.
- The proxy ARP replies with its own MAC address. When the packet arrives, the router delivers it to the appropriate host.





*Note*

**An ARP request is broadcast;  
an ARP reply is unicast.**

## 7.3 RARP

RARP finds the logical address for a machine that only knows its physical address.

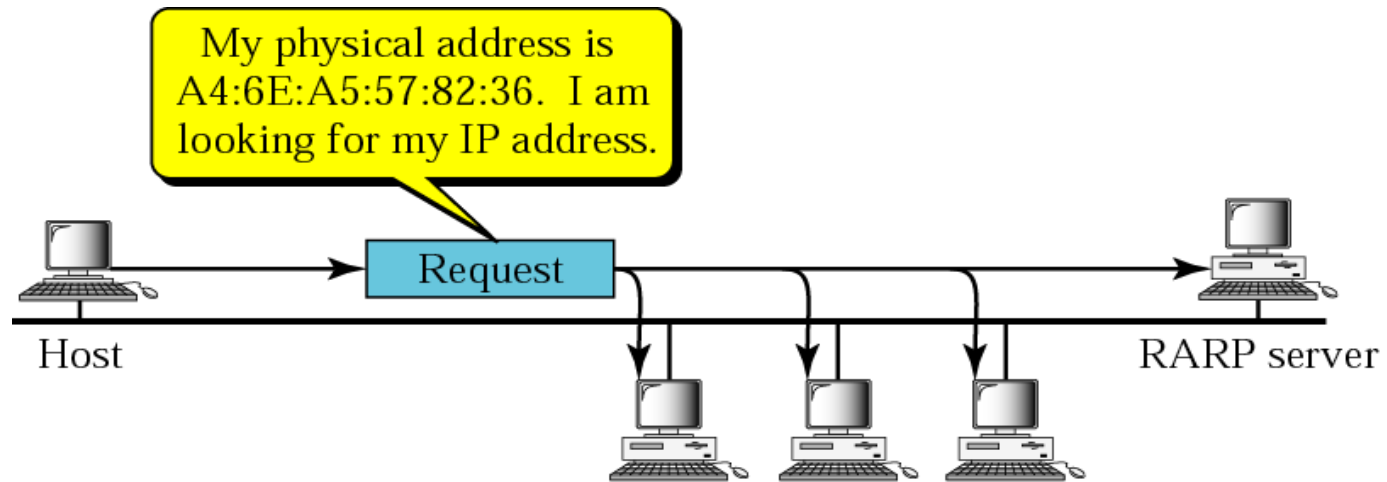
This is often encountered on thin-client workstations. No disk, so when machine is booted, it needs to know its IP address (don't want to burn the IP address into the ROM).

RARP requests are broadcast, RARP replies are unicast.

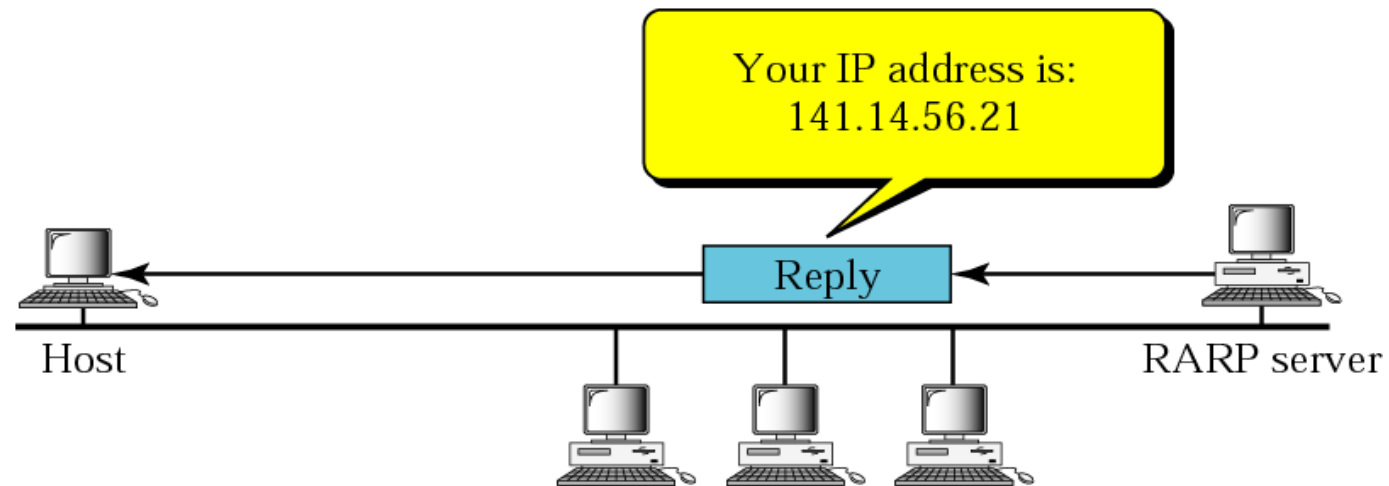
If a thin-client workstation needs to know its IP address, it probably also needs to know its subnet mask, router address, DNS address, etc.

So we need something more than RARP. BOOTP, and now DHCP have replaced RARP.

**Figure 7.10** *RARP operation*



a. RARP request is broadcast



b. RARP reply is unicast

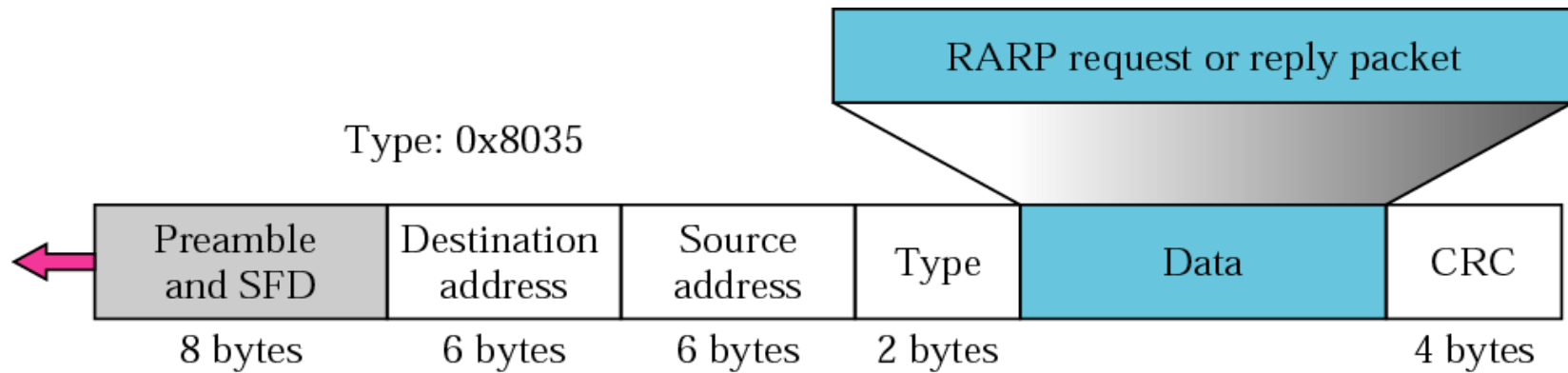


**Figure 7.11** *RARP packet*

| Hardware type                                                                                    |                 | Protocol type                   |
|--------------------------------------------------------------------------------------------------|-----------------|---------------------------------|
| Hardware length                                                                                  | Protocol length | Operation<br>Request 3, Reply 4 |
| Sender hardware address<br>(For example, 6 bytes for Ethernet)                                   |                 |                                 |
| Sender protocol address<br>(For example, 4 bytes for IP)<br>(It is not filled for request)       |                 |                                 |
| Target hardware address<br>(For example, 6 bytes for Ethernet)<br>(It is not filled for request) |                 |                                 |
| Target protocol address<br>(For example, 4 bytes for IP)<br>(It is not filled for request)       |                 |                                 |



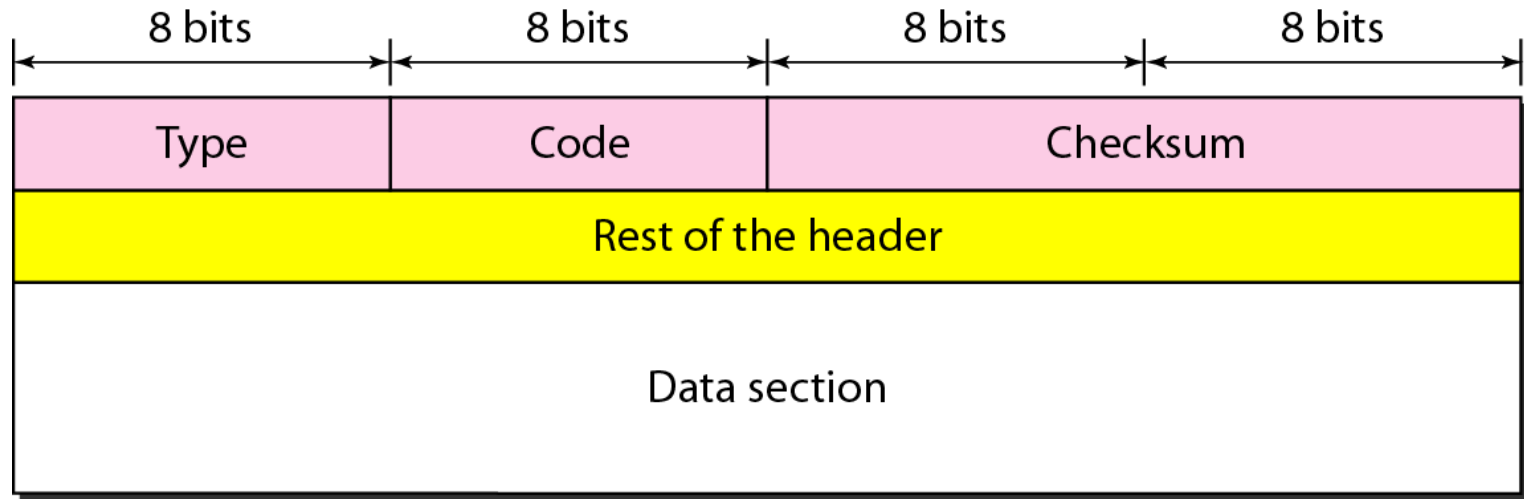
**Figure 7.12** *Encapsulation of RARP packet*



# ICMP

- IP provides unreliable and connectionless datagram delivery.
- The IP protocol has no error-reporting or error-correcting mechanism.
- What happens if router discard a datagram or the time-to-live field has a zero value? What happens if the final destination host must discard all fragments of a datagram? These are examples of situations where an error has occurred and the IP protocol has no built-in mechanism to notify the original host.
- The IP protocol also lacks a mechanism for host and management queries.
- A host sometimes needs to determine if a router or another host is alive. And sometimes a net-work administrator needs information from another host or router.
- The **Internet Control Message Protocol (ICMP)** has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.

**Figure 21.8** *General format of ICMP messages*



- Type field defines the type of the message.
- The code field specifies the reason for the particular message type.
- The code field specifies the reason for the particular message type.



*Note*

**ICMP always reports error messages  
to the original source.**



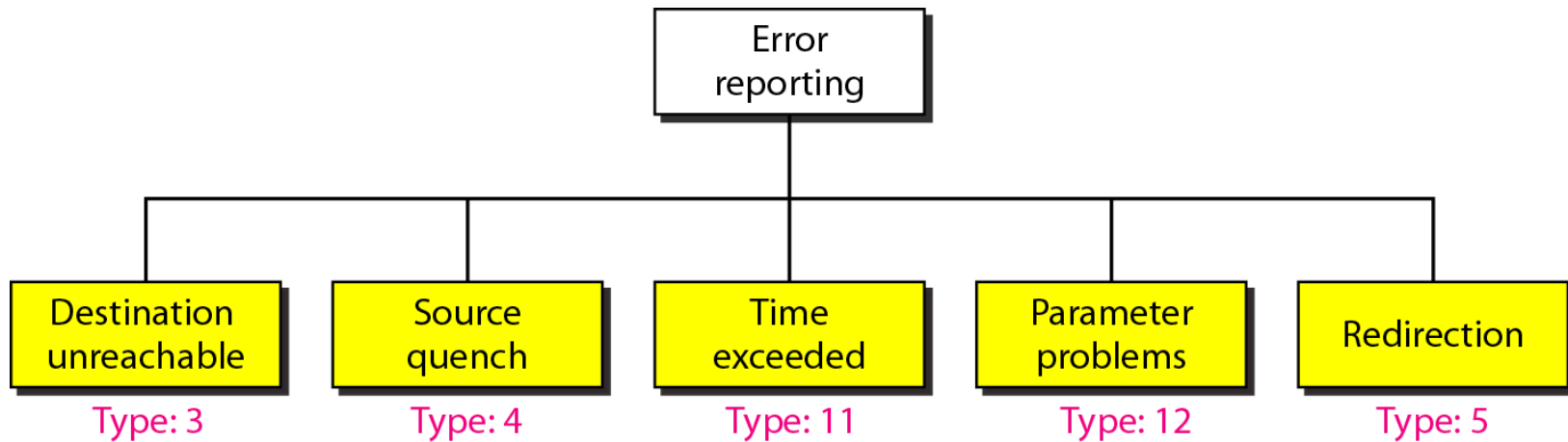
# Type of messages

- **Error Reporting Messages**
- **Query Messages**

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**Figure 21.9** *Error-reporting messages*

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## **Destination Unreachable**

When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram.

## **Source Quench**

When a router or host discards a datagram due to congestion, it sends a source-quench message to the sender of the datagram.

## **Time Exceeded**

When the time-to-live value reaches 0, after decrementing, the router discards the datagram. However, when the datagram is discarded, a time exceeded message must be sent by the router to the original source.

## **Parameter Problem**

If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

## **Redirection**

The router that receives the datagram will forward the datagram to the correct router. To update the routing table of the host, it sends a redirection message to the host.



## **Important points about ICMP error messages:**

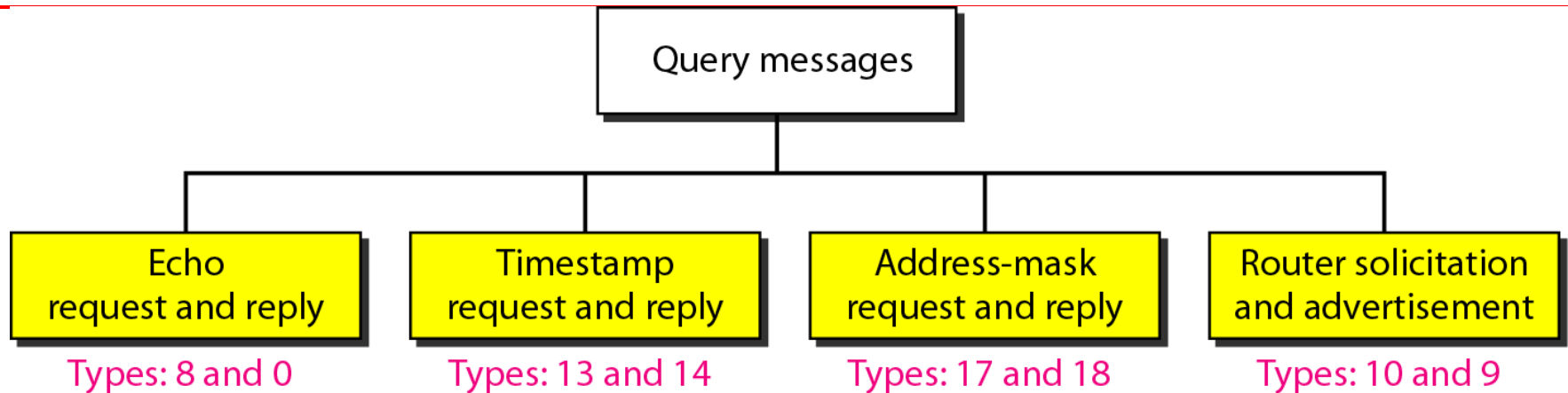
- ❑ No ICMP error message will be generated in response to a datagram carrying an ICMP error message.**
- ❑ No ICMP error message will be generated for a fragmented datagram that is not the first fragment.**
- ❑ No ICMP error message will be generated for a datagram having a multicast address.**
- ❑ No ICMP error message will be generated for a datagram having a special address such as 127.0.0.0 or 0.0.0.0.**



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**Figure 21.12** *Query messages*

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- In addition to error reporting, ICMP can diagnose some network problems with query messages.
- In ICMP message, a node sends a message that is answered in a specific format by the destination node.
- A query message is encapsulated in an IP packet, which in turn is encapsulated in a data link layer frame.

## **Echo Request and Reply**

The echo-request and echo-reply messages are designed for diagnostic purposes. Network managers and users utilize this pair of messages to identify network problems.

## **Timestamp Request and Reply**

Two machines (hosts or routers) can use the timestamp request and timestamp reply messages to determine the round-trip time needed for an IP datagram to travel between them.

## **Address-Mask Request and Reply**

To obtain its mask, a host sends an address-mask-request message to a router on the LAN.

## **Router Solicitation and Advertisement**

A host that wants to send data to a host on another network needs to know the address of routers connected to its own network. The router-solicitation and router-advertisement messages can help in this situation.

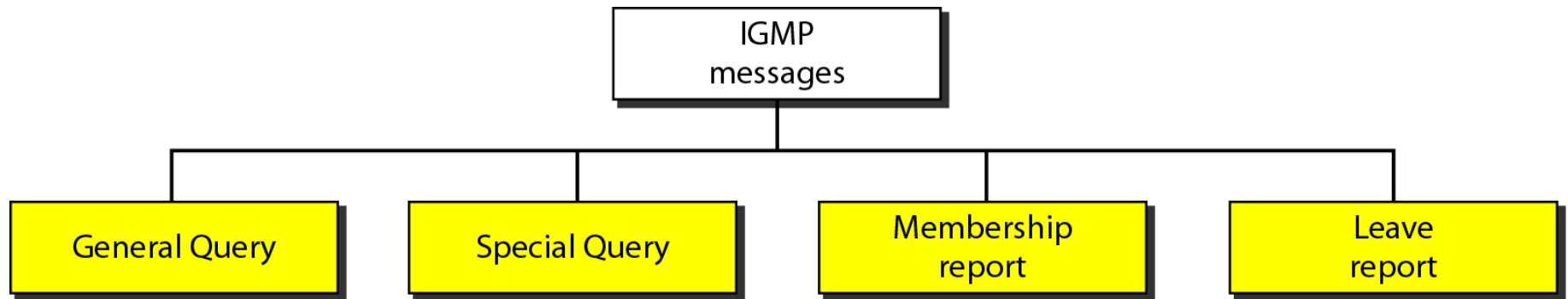
# IGMP

- The IP protocol can be involved in two types of communication: unicasting and multicasting.
- The **Internet Group Management Protocol (IGMP)** is one of the necessary, but not sufficient, protocols that is involved in multicasting. IGMP is a companion to the IP protocol.

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**Figure 21.16** *IGMP message types*

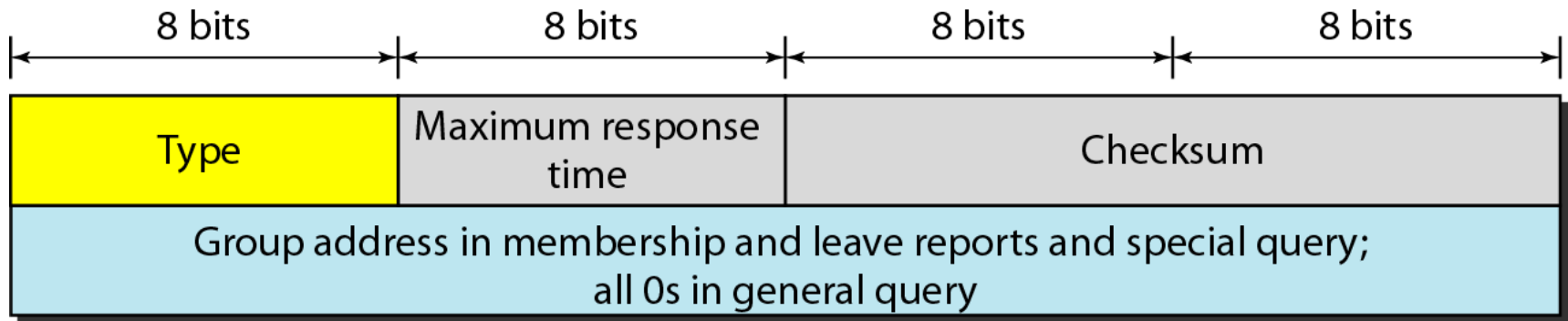
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**Figure 21.17** *IGMP message format*

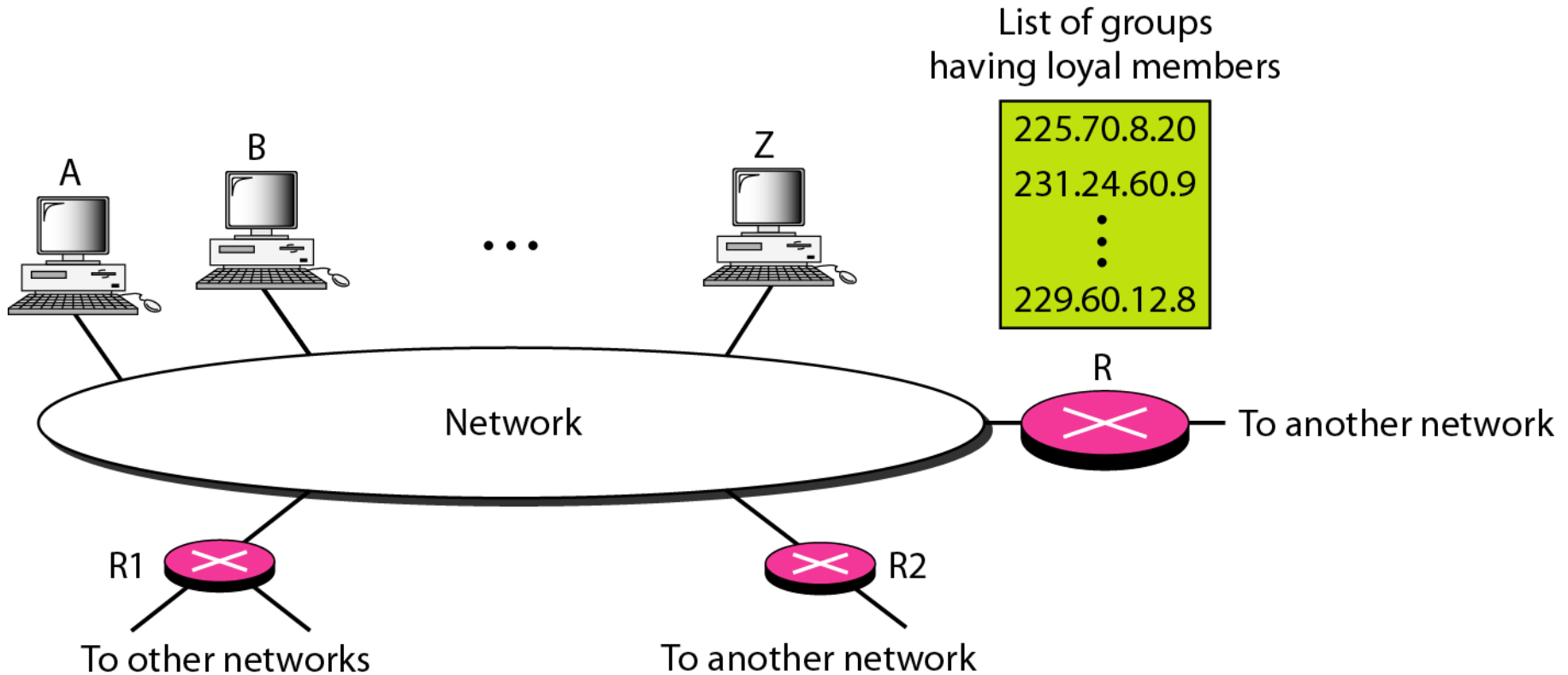
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**Table 21.1** *IGMP type field*

| <i>Type</i>              | <i>Value</i>     |
|--------------------------|------------------|
| General or special query | 0x11 or 00010001 |
| Membership report        | 0x16 or 00010110 |
| Leave report             | 0x17 or 00010111 |

**Figure 21.18** *IGMP operation*



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**Figure 21.20** *Encapsulation of IGMP packet*

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